



Elizabeth Hall, PhD

I specialize in the tech transfer of state-of-the-art research algorithms from prototype to product. I leverage large-scale data analysis, feature engineering, and user behavior and attitudes to improve AI model performance while balancing compute and resource constraints for scalable solutions. At every stage, I bring scientific rigor and a focus on user delight, building high-performance systems that raise the bar for operational excellence at a global scale.

Contact

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Website

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LinkedIn

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Skills

Communication

team leadership
shipping products
data privacy, safety, and policy

Data & Statistics

annotation quality
feature engineering
telemetry dashboards
representational alignment

Generative AI

LLMs/VLMs
neural networks
transformers
embodied avatars (g-splat)
natural language processing
computer vision
electroacoustics

UX

psychometrics
hardware prototypes
AI-mediated reality
AR/VR development
A/B testing
eyetracking

Experience

Meta via Inspyr

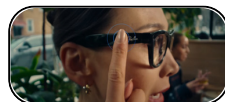
May 2025 - Jan 2026

Hardware UX Researcher

- Led 7 targeted studies quantifying speech enhanced Voice Quality on hardware prototypes; **shared recommendations with leaders and enabled the on-time launch of Conversation Focus at Connect 2025**
- Defined KPI protocols and evaluated e2e latency for real-time chatting on Meta AI glasses over Bluetooth
- **Spearheaded cross-departmental collaboration** to create a VR and AR UX research platform for g-splat Codec Avatars
- **Mentored a team of 4 assistants and interns** to evaluate advanced spatial audio and speech rendering on distributed AR/VR systems with limited compute resources
- **Designed evaluative experiments for all stages of prototype-to-production development**; experience working with tactile transducers, digital sensors, and the Martini sync protocol



Spatial Audio for AR and VR



Conversation Focus with Meta Ray-Ban Glasses

Amazon

May 2023 - Sept 2023

Data Science Intern

- **Developed a LSTM in Python** to predict 10 million customers user behavior patterns with Alexa-brand video/audio devices
- **Communicated key results to C-suite level stakeholders**
- **Reweighted machine learning features** to align forecasts with A/B tests achieving 93% accuracy
- **Built feature engineering pipeline to process millions of customers' data** through Amazon data-storage system improving data processing for a telemetry dashboard

Programming

Languages

Python
R
SQL
MATLAB
C++
C#
Javascript
HTML / CSS
Bash

Machine Learning

pytorch
tensorflow
whisper
torch audio
pandas
openCV
numpy
scipy
fasttext
regressions, ANOVA, t-tests

Data Engineering

Linux
Docker
AWS
S3
ETL
Spark
parallel processing

Education

PhD Psychology

UC Davis 2025

MSc Neuroscience

EHU 2016

BA Liberal Arts

Bennington College 2015

Certifications

via Correlation One:
Data Science (2023)

via Neuromatch Academy:
Comp. Neuroscience (2020)
Deep Learning (2021)
NeuroAI (2024)

DARPA

May 2019 - Sept 2022

Statistical Consultant

- **Statistical consulting** for the University of Melbourne, DARPA Project SCORE, and the Replicats project
- Supported the development and deployment of automated tools that enable DoW personnel to quickly calibrate the validity of behavioral science claims/results



Automated tools to rate research responsibly

UC Davis

Sept 2018 - Dec 2024

Research Fellow

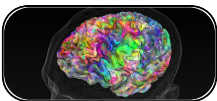
- **Led a 4-year research program** on multisensory attention in VR, with \$180K in funding from the *Office of Naval Research*
- Trained 10 assistants on eyetracking and display calibration
- Developed a psychophysical framework for assessing 3D depth perception and image distortion
- Designed a **computer vision object detection** and eye-tracking systems analysis across multiple VR and AR branded devices
- Created and evaluated **vision-language transformer** models of 30k semantic segmentations **with Pytorch/Tensorflow** to generate textual descriptions of complex, cluttered images
- **Published 7 scientific papers in high-impact journals** and presented at *ICCV, Psychonomics*, and the *Vision Science Society*

NIMH

Aug 2016 - Aug 2018

Research Fellow

- **Designed online experiments** to collect and evaluate human data and semantic segmentations using JavaScript and Amazon Mechanical Turk, collecting data from 2,000 participants
- Recorded 7T BOLD fMRI signal and applied multivariate pattern analyses and neural network based-decoding to identify stimulus perception within the visual cortex and hippocampus
- **Published neural computing and neuroimaging research in** *Nature Communications and Cerebral Cortex*



Meaning in the Brain Changes Ideas on Memory



Our Memory Is Even Better Than We Thought